

EcoFit

Know what the air is really doing to you.

Your fitness tracker knows your heart rate, sleep and strain. It has no idea you spent the afternoon in wildfire smoke. EcoFit is a coin-sized sensor that clips onto the band you already wear — and an app that fuses air quality with your body data.



EcoFit mounts on the underside of the strap — directly opposite the WHOOP sensor. No stacking, no interference, no second device to remember.

The Problem

Air quality apps report your *city*. They don't know your street, your gym, or your bedroom — and they never connect it to how you actually feel.

The Solution

Measure the air at your body, in real time, and overlay it on the fitness data you already trust. One insight neither device could produce alone.

A sensor lab the size of a stick of gum.

36 × 20 × 8 mm. 10-12 grams. Brushed aluminium shell, USB-C charging, and a strap-mount cradle that fits standard fitness bands.

DIMENSIONS & WEIGHT

Length36 mm

Width20 mm

Height8.0 mm

Weight~10-12 grams

TOP VIEW

20 mm

36 mm

EcoFit

SIDE VIEW

8.0 mm

FRONT VIEW

Top, side and front views with dimensions.

PM1 / 2.5 / 10Particulates

VOC / NOxGases

Temp / Humidity

Pressure / Altitude

UV Index

GNSSGPS · Galileo · BeiDou

BLE 5.0Long Range

500 mAhLiPo · USB-C

EcoFit AQI + Fitness Add-on for Whoop

PCB TOP VIEW (Component Side)

36.0 mm

20.0 mm

AIR INTAKE (BMV080)

BMV080 PM SENSOR

BMP390 PRESSURE SENSOR

BATTERY CONNECTOR (JST PH 2-Pin)

BQ25185 CHARGER

SGP41 VOC / NOx

SHT45 TEMP / RH

VEML6075 UV INDEX

nRF52832 QFAA BLE MCU

u-blox MIA-M10Q

TPS62840 3.3V REGULATOR

GNSS ANTENNA CONNECTOR

MIA-M10Q GNSS MODULE

TEST PAD (See List)

BLE ANTENNA (Antenova SR4W089)

PCB BOTTOM VIEW (Solder Side)

TP1 TP2 TP3 TP4 TP5 TP6 TP7 TP8

SWD

1. VDD

2. SWDIO

3. SWCLK

4. RESET

5. GND

KEEP OUT (ALL LAYERS)

GNSS ANTENNA AREA 5.0 x 5.0 mm MIN

KEEP OUT (ALL LAYERS)

BLE ANTENNA AREA 6.0 x 7.6 mm MIN

SIDE PROFILE

8.0 mm

TEST PADS (Top Side)

TP1 - 3V3	TP6 - SCL
TP2 - GND	TP7 - TX (MCU)
TP3 - VBAT	TP8 - RX (MCU)
TP4 - SYS (4.5V)	TP9 - RESET
TP5 - SDA	

MAJOR COMPONENTS

Component	Purpose	Interface
BMV080	PM (PM1/2.5/10)	I ² C
SGP41	VOC / NOx	I ² C
SHT45	Temperature / Humidity	I ² C
BMP390	Pressure / Altitude	I ² C
VEML6075	UV Index	I ² C
nRF52832	MCU / BLE 5.0	I ² C, UART, GPIO, SWD
MIA-M10Q	GNSS (GPS / Galileo / BeiDou / QZSS)	UART
BQ25185	LiPo Charger + Power Path	I ² C
TPS62840	3.3V Buck Regulator	—

PCB STACKUP (4-LAYER)

Layer	Name	Thickness
L1	Top (Signal)	0.018 mm
L2	GND Plane	0.018 mm
L3	3.3V Plane	0.018 mm
L4	Bottom (Signal)	0.018 mm
	Prepreg	0.10 mm
	Core	0.60 mm
	Prepreg	0.10 mm
Total Thickness		~1.00 mm

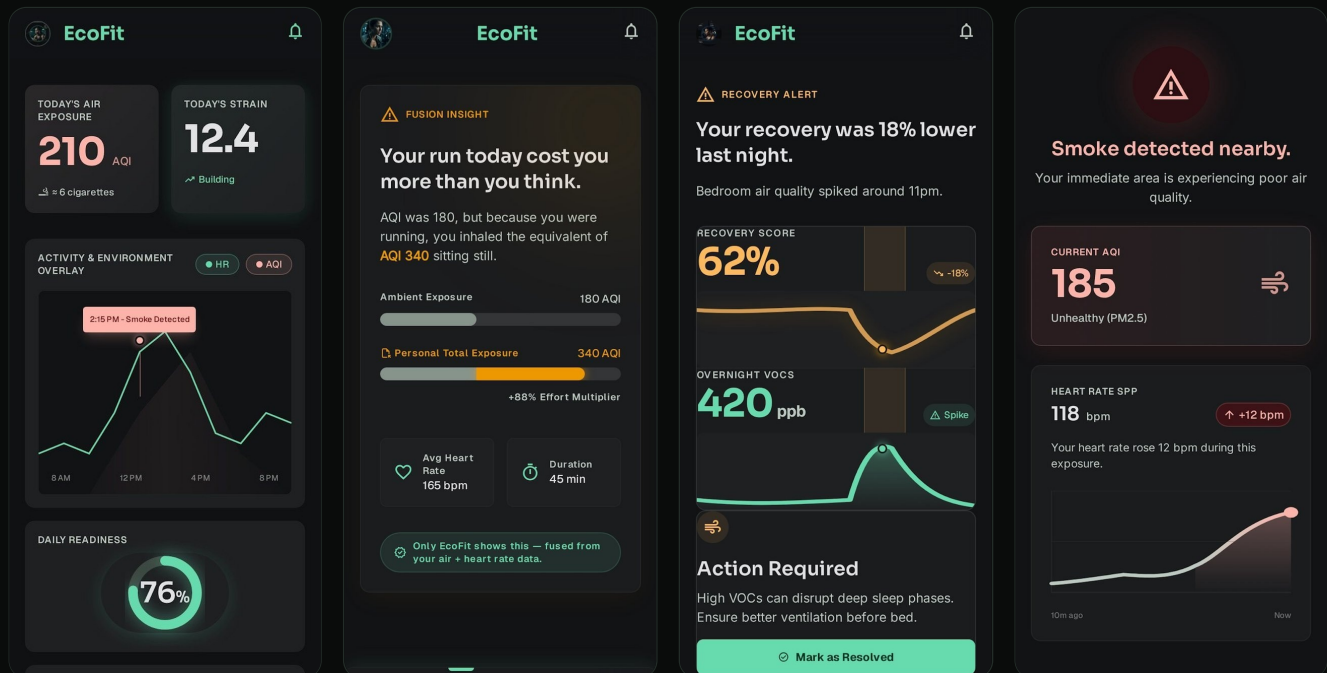
KEY SPECIFICATIONS

- Dimensions: 36.0 x 20.0 x 8.0 mm (Max)
- 4-Layer PCB, 1.0 mm Thick
- Weight (PCB + components): ~10-12 grams
- Power: LiPo 3.7V, 500mAh (typ)
- Charging: USB-C, 5V / 500mA
- BLE: nRF52832 (BLE 5.0 Long Range)
- Sensors: BMV080, SGP41, SHT45, BMP390, VEML6075
- GNSS: u-blox MIA-M10Q
- Operating Temp: -10°C to 55°C
- Storage Temp: -20°C to 60°C

Note: Keepout areas are on all layers as per antenna vendor guidelines.

Four things no other app can tell you.

The sensor is only half the product. The value is in the fusion — what happens when air data and body data sit on the same timeline.



1 · Daily Exposure

Example: "210 AQI today $\approx$ 6 cigarettes." Strain and air on one chart, so a 2:15 PM smoke spike sits right next to your heart rate.

2 · Fusion Insight

Example: "Ambient AQI was 180 — but you were running, so you inhaled the equivalent of 340." Breathing rate changes the dose.

3 · Recovery Alerts

Example: "Recovery was 18% lower last night — bedroom VOCs hit 420 ppb at 11 PM." Then a fix: ventilate before bed.

4 · Live Warnings

Example: "Smoke detected nearby — AQI 185, your heart rate rose 12 bpm." One tap to route toward cleaner air.

"Your run today cost you more than you think." That one sentence *is* the product — and producing it requires a sensor on your body **and** your heart rate. Nobody else has both.

Data people come back for.



Local Leaderboard

Clean Air Score ranked by neighbourhood, built from real member sensors — a live pollution map no public station can match.

Weekly Recap

"Masked 5/7 days. Avoided 3 smoke events. Recovery up 8%." Shareable, and the reason the streak keeps going.

Wear It Anywhere

Wrist, necklace, bag, belt or shirt. The app recalibrates for placement, so the reading stays honest.

30-Second Setup

Pair over Bluetooth, connect your existing fitness account, done. No new ecosystem to adopt.

1

Add-on, not replacement

We don't compete with WHOOP, Garmin or Apple Watch. We attach to them — and make their data more valuable.

2

A network effect

Every device improves the map. More members means finer-grained air data for everyone in the city.

3

Timing

Wildfire smoke, indoor air and long-term exposure have moved from niche worry to daily reality.

Where we are: hardware design and PCB complete, app flows designed, wearable form factor validated. Next up: prototype build and pilot testing.